

Bileaflet Mechanical Heart Valves

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ABSTRACT

This study highlights the scientific innovation and transformative impact of developing bi-leaflet prosthetic heart valves, marking a significant advancement in cardiac surgery and patient care. Materials were selected based on biocompatibility, thromboresistance, hemodynamic efficiency, and durability, prioritizing metal alloys such as titanium and pyrolytic carbon, alongside synthetic polymers like Polytetrafluoroethylene (PTFE) fabric. Manufacturing methods emphasized accuracy, focusing on surface modification and coating technologies, with techniques such as precision laser cutting and vapor deposition. Patient preferences revealed a prioritization of hemodynamic performance and risk of secondary interventions over other factors, shaping the bi-leaflet valve design to include a 90° open angle for optimized blood flow. Secondary considerations, such as thrombogenicity and smooth surface properties, were also integrated to enhance overall functionality.

KEYWORDS

Mechanical Valve, Bileaflet Design, Anticoagulation, Polytetrafluoroethylene, Pyrolytic Carbon.

1. INTRODUCTION

1.1 Physiology of Heart Valves

Heart valves (HVs) are specialized structures within the heart that ensure unidirectional blood flow throughout the cardiac cycle. The semilunar pulmonary (PV) and aortic (AV) valves prevent retrograde flow back into the ventricles during diastole, while the atrioventricular tricuspid (TV) and mitral (MV) valves restrict reverse flow from the ventricle to the atrium during systole. Collectively, HVs must perform their functions approximately 3 billion times over an average human lifetime for successful cardiac homeostasis and overall circulatory health.^[1,2] Naturally, HVs' structure can be compromised due to degeneration over time caused by diseases (e.g., degenerative calcific stenosis and infective endocarditis) or tissue aging that affect HVs' precision for maintaining circulatory efficiency. When HV malfunctions occur, the consequences for cardiac function can be severe if not treated appropriately. HVs' malfunctions often require medical or surgical intervention, leading to the replacement of damaged valves with prosthetic heart valves (PHVs).

1.2 Scientific Innovation of Prosthetic Heart Valves

Prosthetic heart valves (PHVs) are artificially engineered valves that mimic the function of HVs (**Figure 1**). The introduction of PHVs represented a significant advancement in cardiac surgery, offering a viable solution for patients with compromised HVs. Modern PHVs can be categorized into mechanical and biological valves. Mechanical valves, constructed from durable materials (e.g., carbon and titanium) offer longevity, while biological valves, derived from porcine or bovine tissues, provide a more natural hemodynamic profile.^[1] The ideal PHV should mimic the properties of a native heart valve in hemodynamics, durability, thromboresistance, and implantability.

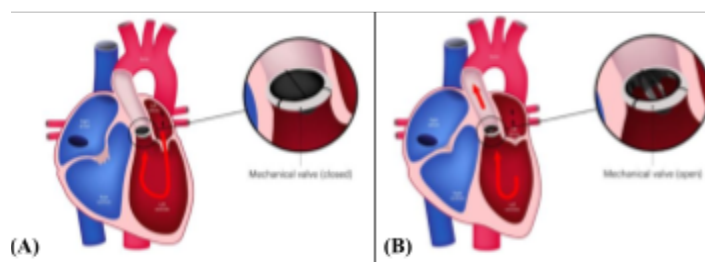


Figure 1. Blood circulation with mechanical aortic valve (A) MV opened during diastole. (B) MV closed during systole.

1.3 Market Value & Motivation

The global prosthetic heart valve market was valued at \$8.6 billion in 2023, and it is projected to reach \$30.3 billion by 2035, reflecting a compound annual growth rate (CAGR) of 11.1% from

2024 to 2035.^[2] This growth is primarily driven by the increasing incidence of heart valve replacements and the rising prevalence of heart diseases globally. The rise in heart valve replacement procedures can be attributed to advancements in surgical techniques and innovative prosthetic designs. Improving patient quality of life (QoL) is a crucial factor driving the market, as healthcare providers increasingly recognize the importance of effective treatment options for heart valve diseases.^[1,2]

The development of accessible and affordable mechanical valve prostheses is critical to combating health disparities.^[1] By developing cost-effective solutions, we can break down barriers that prevent access to life-saving treatments for those in need. Furthermore, our proposed design will reduce reliance on scarce biomaterials and address biocompatibility issues, ensuring increased availability and patient QOL.^[3] Ultimately, our goal is to develop an affordable, durable, and innovative mitral valve to ensure access in underdeveloped regions.

1.4 Previous Biomedical Device Development

Before bileaflet valves were developed, caged ball valves and tilting disc valves were used. The caged ball valves were designed with a silicone or metal ball enclosed in a metal cage. This design failed due to the inefficiency in mimicking natural blood flow, often leading to blood stagnation in certain areas which increased the risk of blood clot formation or thromboembolism.

Next, tilting disk valves were developed, which were composed of a single circular disk that pivots around an axis. To allow for blood flow, the disk would tilt; and to prevent backflow the disk would close. This was an improvement from the caged ball valve as it is more durable due to there being fewer moving parts. Despite having better blood flow, the tilting disk still blocked part of the orifice and increased the risk of thromboembolism.^[4] To combat the limitations in previous designs, bileaflet valves consisting of two semicircular leaflets were developed to properly mimic the function of a natural heart valve.^[5]

1.5 Research Gaps

Although bileaflet valves improved blood flow, patients continue to require lifelong anticoagulation treatment as the risk of thromboembolism is persistent due to the non-physiological blood flow patterns around the device. This blood flow pattern also leads to turbulence and increased stress on red blood cells which results in platelet activation. In order to combat these issues, more research is required on the interaction between blood and device materials. Patient-specific valve designs that are tailored to an individual's unique anatomy could improve blood flow. Currently, mechanical heart valves are a one-size-fits-all solution which is inefficient due a plethora of differences in anatomical characteristics such as the size of an individual's heart, the positioning of the valve, and blood type.^[6]

Bileaflet valves are durable, lasting approximately 20 to 30 years; however, with long-term use the device is prone to structural fatigue and wear in the leaflet pivots or hinges that can lead to mechanical failure. Research must be conducted under mimicked real-life conditions to determine which materials can withstand continuous and high stress environments in the heart. Materials need to be rigorously tested to ensure they are durable and can perform reliably over decades of use.^[1]

2. METHODS

2.1 Design Scheme for BMHV

To complete a design of a Bileaflet Mechanical Heart Valve (BMHV), optimal functionality within the cardiovascular system is the first and foremost factor to consider. A basic schematic BMHV composed of leaflets, valve housing and pivot mechanism was designed in SolidWorks, a

3D CAD program. In order to improve on the hemodynamic performance, the leaflets were mechanically designed to open at 90 degrees to reduce turbulent flow. In addition to the mechanical design, the selection of materials were considered for each part of the valve which included the mechanical parts mentioned above and the sewing ring. Additionally, functional requirements (durability, availability for multiple sizes), the feasibility of manufacturing technology, and the regulatory compliance will be discussed.

2.2 Materials Selection for BMHV

2.2.1 Material Selection for the Leaflets

The requirements considered for the leaflet material are high strength, enough durability, low thrombogenic properties and a smooth surface. Pyrolytic Carbon (PyC) is the mainstream choice for the leaflets, which meets Some designs aimed to combine leaflets with other chemical additives, such as silicon; however, full PyC designs have shown to exhibit higher levels of hydrophobicity, resulting in strong thromboresistance. Therefore, leaflets of BMHV would be composed of pure PyC.

2.2.2 Material Selection for the Valve Housing and Pivot

As high durability and mechanical strength is desired, these parts tend to be made from different alloys and also PyC. However, PyC is more brittle than alloys.^[7] In order to combine the thromboresistant advantage of PyC with the advantages of alloys in durability and corrosion resistance, the valve housing and pivot parts of BMHV would be composed of titanium alloy with PyC coating.

2.2.3 Material Selection for the Sewing Ring

Since this part is directly attached to the patient's tissue, the biocompatibility and the security considerations are critical. This material should also provide flexibility for surgical implantation. Polyester fabric, Polyurethane and silicone are the most common choices. Here, referring to the design of On-X Prosthetic Heart Valve, the sewing ring would be constructed of polytetrafluoroethylene (PTFE) fabric.^[8]

2.3 Manufacturing Technology for BMHV

2.3.1 Surface Modification and Coating

Surface modification is used to address complications to reduce inflammation, promote tissue integration, enhance long-term durability, and minimize the risk of infection. Various physical and chemical methods can be employed to modify the polymer surfaces of heart valves, including plasma treatment, ion implantation, and UV irradiation.^[8] Coating technology is crucial to meet the specific requirements of the blood-contacting biomaterial, such as the intended use, duration of exposure to blood, and desired performance characteristics. Nanosecond laser etching technology can be used to create a superhydrophobic surface, which is important to minimize interactions between blood and materials, thus enhancing anticoagulation and inhibiting thrombus formation. Nanosecond laser etching technology can be used to create the coating.^[9]

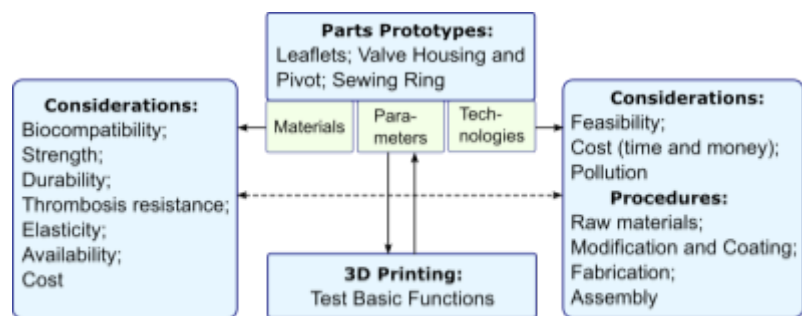


Figure 2. The roadmap for design scheme, materials selection and manufacturing technology.

2.3.2 Fabrication of Components

For the leaflets made of PyC, physical vapor deposition (PVD) or chemical vapor deposition (CVD) are two commonly used technologies, providing a hard, smooth, and chemically inert surface. As to other mechanical parts, precision laser cutting or electrical discharge machining would be firstly used to shape the materials. Next, the thermal process would be conducted to achieve the detail parts. Finally, electro-polishing would be applied to ensure a smooth surface across all parts.

3. RESULTS AND DISCUSSION

3.1 Design Benefits and QFD Chart

Based on a clinical study performed in 2015 among 132 patients, a detailed value of the importance of risks and benefits associated with the mechanical HV were obtained.^[10] Using the results from this study, the customer importance rating was concluded in addition to supplemental factors which are outlined in a quality function deployment (QFD) chart which is depicted in Fig. 3. What the patients deem as the most critical factor is the probability for a second valve operation. Technical requirements related to the mechanical HV and the relationship between the customer and technical requirements are provided in the QFD chart.

The most significant technical requirement was achieving a hemodynamic performance that almost imitates a natural valve through the technical importance score. The hemodynamic performance was specifically divided into 3 most important mechanical aspects. Our design to make the leaflets have a 90° open angle aims to achieve less turbulent flow.^[11]

Customer importance rating	Technical requirements Customer requirements	Hemodynamic Performance			Thrombosis Resistance	Biocompatible Material	Durability	Weighted Score
		Turbulent flow	Flow Velocity Profile	Effective Orifice Area				
3	Possible bleeding	3	2	1		6	3	45
2	Blood clots	1	2	3	9			30
5	Second valve operation	4	1	1	6	3	9	120
3	Limitations of daily activities	4	4	1	3			36
4	Valve sound	4	3	2			3	48
1	Cost	2	0	1	9	6	6	24
	Technical importance score	61	39	26	66	39	72	

Relationships: Strong 9 Moderate 6 Weak 3 None 0

Figure 3. Quality Function Deployment (QFD) Chart

3.2 Material Benefits

Referring to the selected materials aforementioned, leaflets made of PyC can provide high durability, low thrombogenicity and a smooth surface. The valve housing and pivot parts made of a titanium alloy with PyC coating are both corrosion resistant and thromboresistant. The sewing ring made of PTFE fabric is both a secure interface to reduce possible bleeding and a highly flexible material for surgical implantation. In summary, the materials in our design can achieve both basic requirements (e.g. biocompatibility, durability and mechanical strength) and superior requirements (e.g. thromboresistance and flexibility).^[11]

3.3 Technical Advantages

The manufacturing technologies employed for the BMHV components collectively address critical design and functional requirements, ensuring the device meets clinical and

Critical part requirements	Leaflets	Housing	Sewing ring	Specifications
Dimensional precision	•	• •	•	±0.01 mm
Surface smoothness	• •	• • •	•	Nano-level
Material strength	• • •	• • •	•	>100 MPa
Biocompatibility	• • •	• •	•	ISO 10993
Durability	• • •	• • •	• • •	>10 years

Figure 4. Part Planning Matrix

performance standards. The matrix depicted in Fig. 4, demonstrates how these technologies provide distinct advantages. Precision engineering techniques, including chemical vapor deposition, laser cutting, and nano-level surface treatments, deliver exceptional dimensional accuracy and smooth surfaces, ensuring superior hemodynamic performance and reduced flow turbulence.^[10, 11]

3.4 Technical Figure to Summarize Key Results

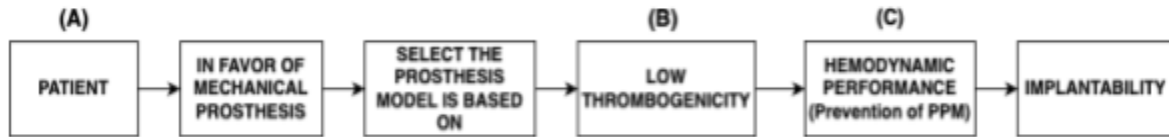


Figure 5: Algorithm for the selection of the optimal prosthesis in the individual patient. (A) Initial patient conditions (i.e. age, life expectancy, preference) are considered to determine whether a mechanical heart prosthesis would be effective. (B) Next, thrombogenicity conditions are considered. Low thrombogenicity is favorable. (C) Finally hemodynamic performance (explained above) and implantability are considered. These conditions are patient-dependent and are affected by body size, valve size, age, lifestyle, comorbidities, and left ventricular function.^[12]

4. PROTOTYPE DESIGN AND 3D PRINTING TRIAL

4.1 Early considerations

Our initial approach to designing our own BMHV was to create a two-part assembly consisting of two “leaflet” parts and a “casing” part. The casing was designed with two holes on each side, allowing the separately printed leaflets to be securely “popped” into place by the pegs on the leaflet. In addition to considering the structural design of the BMHV, we also desired to ensure that we could achieve a fundamental function of HVs: allowing fluid to enter in one direction, while preventing backflow in the opposite direction.

4.2 Failed Trials and Revisions

To achieve the first prototype of our BMHV, we designed a thick casing part with a small-scale lip inside. That way, the leaflets could rest on top of the small lip, preventing backflow in the opposite direction. The leaflets and the casing were printed separately. We decided to print our first prototype with a ratio of 4.23:1 ratio with respect to the size of mitral heart valves (normally 30 mm in diameter.^[13]) This way, any errors in our larger prototype can be easily identified. Initially, the pegs on the leaflet were short, just long enough to be inserted into the holes on the casing. Although the holes in the casing were designed to accommodate the leaflet pegs, the 3D printing process introduced constraints due to the larger diameter printing filament. This unexpectedly created larger pegs and smaller casing holes which led to hindrances regarding the insertion of the leaflets into the casing. Fortunately, with hot glue and metal wires, the parts in this print were salvaged and a prototype was made. Although we still wanted to ensure 3D printability, this gave us a better idea on how to approach revisions to overcome printing issues.

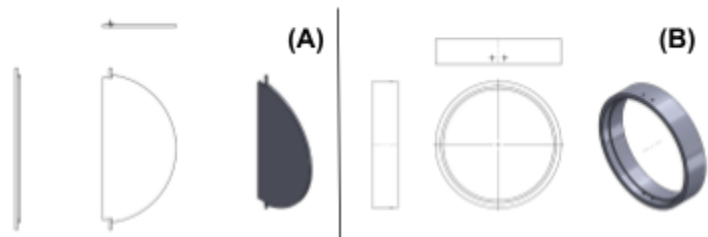


Figure 6: (A) Solidworks design for first printing trial of artificial heart valve leaflet part. The entire length of the leaflet was 127 mm. (B) Solidworks design for first printing trial of artificial heart valve casing part. The outer diameter of the casing was 127 mm.

In our second printing trial, we decided to print both the leaflets and the casing simultaneously and kept the same dimensions as our first printing trial. This was done to avoid any errors associated with inserting the leaflets into the casing as the leaflets will already be in the correct location when printing is concluded. However, when the print finished, the leaflet pegs fused with the casing holes during the printing process. This caused the leaflets to be stiff and immovable, which is not ideal for our device's mechanism that needs to continuously open and close.



Figure 7: Fusion of leaflet pegs and casing holes in second printing trial

4.3 Final Design

To ensure that the pegs of the leaflets were not going to fuse with the holes in the casing, we redesigned the casing so that it possessed larger holes and redesigned the leaflets so that the pegs were longer. This redesign ensured that removable scaffolding could be printed between the leaflet pegs and the casing hole. To reflect the actual size of the mitral valve (nearly 1:1 ratio), the design was scaled down to possess an outer diameter of 28 mm. After printing, the scaffolding was easily removable, allowing for the leaflets to move freely. Finally, both the structural and functional considerations were implemented so that the valve opened when pressure was applied from behind, and closed when pressure was applied in the opposite direction – all achieved through 3D printing.



Figure 8: Final 3D-printed Bileaflet Mechanical Heart Valve Prototype

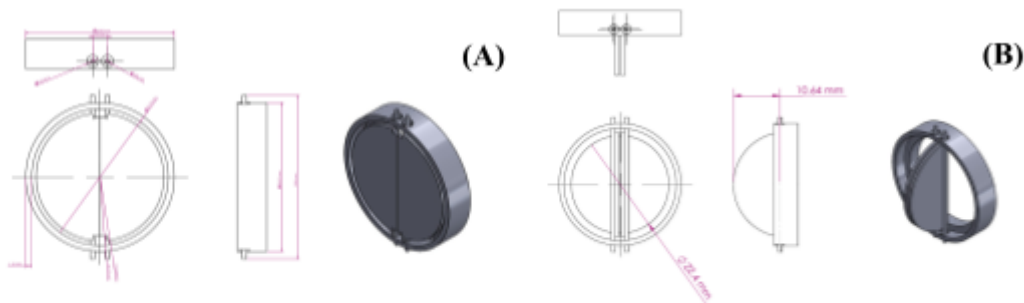


Figure 9. The final design for the Bileaflet Mechanical Heart Valve. **(A)** Closed position of the valve. Includes measurements. **(B)** Open position of the valve. Includes measurements.

5. CONCLUSION

The field of 3D-printed biomedical implants is rapidly advancing, and as researchers start to understand the factors influencing the successful functionality of 3D-printed heart valves, this technology has the potential to offer personalized treatment that is tailored to an individual patient's needs. More durable, mechanically optimal, biocompatible, and easily implanted designs are necessary to address the widespread clinical need for this technology.

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